Spatiotemporal periodicity of dislocation dynamics in a two-dimensional microfluidic crystal flowing in a tapered channel

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When a many-body system is driven away from equilibrium, order can spontaneously emerge in places where disorder might be expected. Here we report an unexpected order in the flow of a concentrated emulsion in a tapered microfluidic channel. The velocity profiles of individual drops in the emulsion show periodic patterns in both space and time. Such periodic patterns appear surprising from both a fluid and a solid mechanics point of view. In particular, when the emulsion is considered as a soft crystal under extrusion, a disordered scenario might be expected based on the stochastic nature of dislocation dynamics in microscopic crystals. However, an orchestrated sequence of dislocation nucleation and migration is observed to give rise to a highly ordered deformation mode. This discovery suggests that nanocrystals can be made to deform more controllably than previously thought. It can also lead to novel flow control and mixing strategies in droplet microfluidics.

many-body system | order | microfluidic crystal | dislocation dynamics | plasticity

The emergence of order and spontaneous self-organization have been of long-standing interest (1). They are key not only to the understanding of complex phenomena from chemical oscillators to swarming behavior in animals (2, 3), but also to novel engineering solutions if harnessed (4, 5). Two-phase flow in microfluidics offers a simple platform for the study of dissipative nonequilibrium systems, where hydrodynamic interactions have led to the emergence of collective dynamics and order (6–10). Most works thus far have focused on dilute emulsions or foams in simple channel geometries. Whereas rich physics has been revealed, these phenomena have yet to find implications in the broader technological context. Nevertheless, concentrated emulsions, bubble rafts, and colloids have long been used as models of crystals for studying grain boundaries, dislocations, plasticity, and other processes central to materials science and solid mechanics (11–16). Such systems have not been applied to model the deformation modes of nanocrystals, however. Recent work on crystal plasticity in microscopic samples found that in contrast to their macroscopic counterparts, both the external geometry and internal structure of the material determine material strength (17). Furthermore, the size and timing of dislocation-induced strain bursts are found to be intermittent, stochastic, and unpredictable (17–20). The stochastic nature of dislocation dynamics complicates the control of the shape of the materials during deformation, and renders their subsequent manipulation and manufacturing challenging (18). What is unknown, however, is whether plasticity remains stochastic as the sample shrinks to the nanoscale, and whether the dimensionality and loading conditions influence the stochasticity. Given the increasing importance of low-dimensional nanodevices in applications from optoelectronics to energy conversion, it is critical to understand how nanomaterials can be shaped, manipulated, and manufactured.

In this paper, we describe an unexpected order in the flow of a quasi-2D concentrated emulsion in a tapered microfluidic channel. We show that the phenomenon can be explained by treating the system as crystal plasticity in the nanoscale, where the size of the extruder is on the order of a few to tens of atoms as modeled by droplets. The confinement of the 2D soft crystal in the extrusion flow causes the reorganization of the crystal's internal structure in a highly ordered manner. The process is robust, and occurs almost indefinitely. Our results represent a type of collective order that could find practical use in the design of flow control systems for effective on-chip droplet manipulations such as mixing and incubation (21, 22). In addition, the phenomenon here directly contrasts the stochasticity of dislocations in microscopic crystals, and implies that the mechanical forming of confined low-dimensional materials could be more easily controlled than previously anticipated.

Results and Discussions

Plug-Like Time-Averaged Velocity Profile. Fig. 1*A* shows the tapered microchannel into which the concentrated emulsion was injected. The taper had a half-angle of 5° leading to a constriction with a width of 30 μm. The emulsion consisted of monodisperse 50-pL water droplets at a volume fraction of ∼85% suspended in a fluorinated oil (*Materials and Methods*). The drops flowed as a 2D monolayer as the channel height was less than one droplet diameter. As they flowed, the drops arranged into a hexagonally packed crystal (Fig. 1*A*) (23). Due to the tapered channel geometry, the emulsion experienced gradual, elastic compression in the transverse direction as it moved along the channel, except at specific locations where the number of rows of drops N (counted at 60° relative to the x axis across the width of the channel) decreased by one (Fig. 1*B*). These locations, referred to as rearrangement

Significance

Collective interactions in many-body systems can give rise to unexpected order. Such interactions underlie a wide range of complex phenomena such as swarming in animals and traffic patterns. This work reports an unexpected order in the flow of highly confined and highly concentrated water-in-oil drops, which can be explained and modeled as a soft crystal being extruded in the nanoscale. The findings are important to the understanding of out-of-equilibrium many-body systems, and the flow control of these drops used as microreactors in droplet microfluidics. Furthermore, contrary to the unpredictable microscale crystal deformation process, the discoveries here indicate that nanoscale crystal deformation can be highly ordered and predictable, and imply that the manufacturing of nanocrystalline materials may be easier than perceived.

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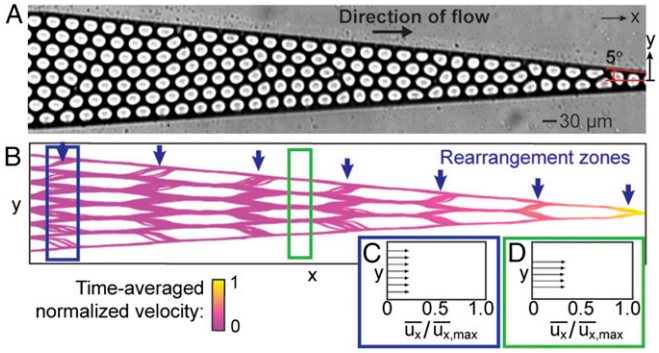

Fig. 1. Channel geometry and time-averaged flow profile. (*A*) Microscope image of the concentrated emulsion in the tapered microfluidic channel. (*B*) The time-averaged total velocity of drops at different locations in the channel. The velocity was normalized to the maximum total velocity measured at the constriction. The position of each marker represents the position where the droplet velocities were averaged. The color of the marker represents the magnitude of droplet velocity at that location. Note that the markers overlap and appear as continuous lines which reflect the trajectories of the drops. The rearrangement zones are indicated by blue arrows. (*C* and *D*) Time-averaged *x* component of droplet velocity $\overline{u_x}$ across the width of the channel measured at the locations marked by the blue and green boxes in *B*. The velocities shown were normalized to $\overline{u_{x,max}}$, the maximum *x* component of velocity measured in the constriction.

zones, occurred at well-defined regions and were approximately equally spaced. We focus on the slow flow regime where the Reynolds number was $Re \sim 10^{-1}$ and the capillary number was $Ca \sim 10^{-4}$ measured at the constriction. Under these conditions, the time-averaged velocity profile of the drops at a given *x* position was uniform or plug-like (Fig. 1 *C* and *D*), consistent with previous reports on concentrated emulsions flowing in channels with constant widths (22).

Unexpected Periodicity in Instantaneous Velocity Profile. The instantaneous velocity of individual drops reveals unexpected periodicity in both space and time. Fig. 2*A* shows a kymograph of the magnitude of instantaneous droplet velocity at different *y* positions in a rearrangement zone (N = 7 to 6) as a function of time. The kymograph indicates that, at a given time, there was always a drop that moved faster than the others within the rearrangement zone. The *y* position of the fast-moving drop shuttled periodically between the top and bottom walls. The same trend was observed in the kymograph in the nonrearrangement zone (N = 7), except it was the position of the slow-moving drop that oscillated between the walls (Fig. 2*B*). Similar behavior occurred in other regions of the channel. Furthermore, the directions of the instantaneous velocity vectors of the drops were counterintuitive from a fluid mechanics point of view (Fig. 2*C*). For a simple liquid flowing in a converging channel, the velocity vectors should point downward in the upper half of the channel, and upward in the lower half of the channel. Here we observed velocity vectors that pointed upward in the upper half of the channel and vice versa (also see Movie S1).

T1 Cascades. Close examination of droplet dynamics shows that the anomalous velocity profiles originated from the rearrangement zones, and were the result of a cascade of T1 events. A T1 is an elementary rearrangement process involving the exchange of neighbors among four drops (Fig. 3*A*), consisting of one pair of diverging drops (drops 2 and 4) and one pair of converging drops (drops 1 and 3) (24, 25). Within each rearrangement zone, there was one T1 occurring at a time. The T1 propagated along a single direction (+60° or −60° relative to the *x* axis) until it reached the wall, where it became "reflected" and propagated toward the opposing wall (along the −60° or +60° direction). This shuttling of T1 between the upper and lower walls repeated itself at a nearly constant velocity in a remarkably regular manner (Movie S2). The periodic patterns in the kymographs (Fig. 2) resulted directly from the periodic T1 motions: We observed that the diverging drop downstream (drop 4) always moved faster than the others (Movie S3). This fast-moving drop retarded the drops immediately upstream. As the retardation propagated to the nonrearrangement zone upstream, a similar periodic pattern mirroring that in the rearrangement zone was formed.

Analogy to Crystal Plasticity During Extrusion. These experimental observations naturally lead to the following questions: (*i*) Why are the rearrangement zones periodically arranged, and what determines this periodicity in space? (*ii*) Why do the T1 cascades occur regularly and repeatedly, and what determines this periodicity in time? (*iii*) What are the conditions necessary for this highly ordered dynamics to emerge, and what will destroy this order? We show that these questions could be answered by treating the concentrated emulsion as a 2D soft crystal, and by applying the mechanics of crystal dislocations.

The T1 cascade is equivalent to the glide motion of a crystal dislocation on its slip plane, where the location of the T1 corresponds to that of a crystal dislocation. The Burgers vector of a dislocation can be determined from the Burgers circuit (Fig. 3 *B* and *C*). A Burgers circuit is obtained by jumping between neighboring drops in a sequence of steps that would form a closed counterclockwise loop in a perfect crystal (26). If the circuit encloses a dislocation, the loop is not closed. The vector pointing from the starting point (S) to the end point (F) of the loop is the Burgers vector of the dislocation. In Fig. 3*B*, the Burgers vector is at 60° from the *x* axis. The corresponding dislocation in a 3D crystal would be called an edge dislocation whose line direction points out of the plane. The dislocation is able to glide on its slip plane, which contains the Burgers vector. Dislocation motion is driven by the resolved shear stress on the slip plane, and is caused by the greater compressive stress in the *y* direction than in the *x* direction (26). Fig. 3 *B* and *C* shows the directions of motion relative to the crystal for the two types of dislocations in our system. In both cases, the *x* component of the velocities was negative. This negative velocity in *x* was canceled, however, by the motion of the entire crystal (extrusion) in the positive *x* direction. As such, the dislocation was almost stationary

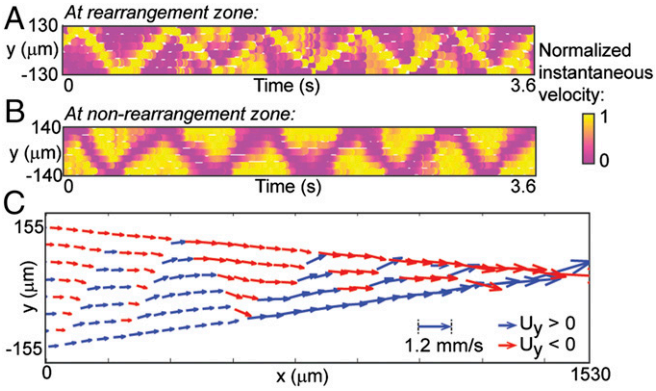

Fig. 2. Unexpected periodicity in instantaneous velocity profile. (*A* and *B*) Kymographs of instantaneous droplet velocities in the rearrangement zone (N = 7 to 6), and nonrearrangement zone (N = 7), respectively. *N* is the number of rows of drops across channel width. The position (*t*, *y*) of each marker represents the time *t* and the *y* position of a drop in the channel when the velocity of the drop was measured. The color of the marker represents the magnitude of the total velocity of the drop normalized to the maximum velocity measured at the constriction. (*C*) Instantaneous velocity vectors of the drops in the channel. Also see Movie S1.

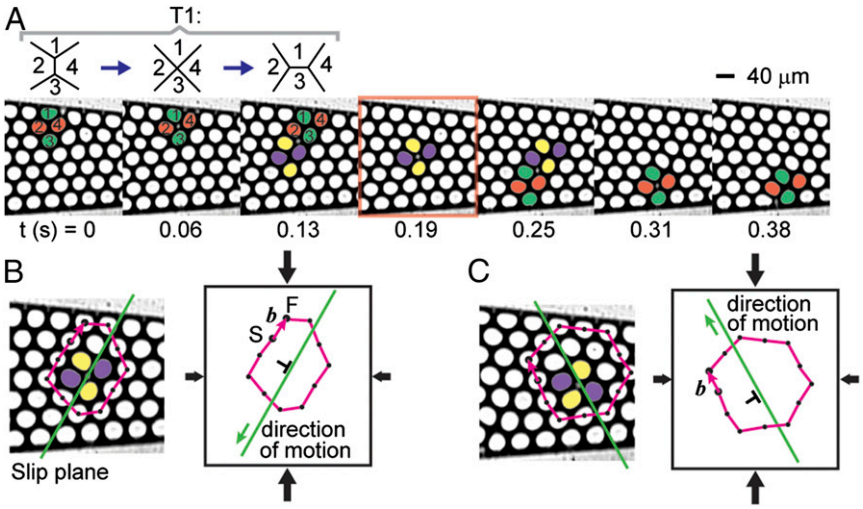

Fig. 3. T1 cascades and Burgers circuit. (*A*) A T1 event involves one pair of diverging drops (2 and 4) and one pair of converging drops (1 and 3). Snapshots of the T1 cascade which is equivalent to a dislocation glide. Three T1 events were highlighted: event 1 (t = 0–0.13 s) where the diverging (converging) drops were colored in red (green); event 2 (t = 0.13–0.25 s) where the diverging (converging) drops were colored in purple (yellow); event 3 (t = 0.25–0.38 s) where the diverging (converging) drops were colored in red (green). (*B* and *C*) Burgers circuits (pink lines) of the dislocations in the crystal at t = 0.19 s (also indicated by the red box in *A*, and at t = 0.58 s, respectively. The pink arrow indicates the Burgers vector $\boldsymbol{b}$. The green line indicates the slip plane, and the green arrow indicates the direction of motion of the dislocation. The black arrows indicate the stress acting on the crystal.

in the x direction relative to the laboratory coordinate system, and only appeared to be shuttling up and down (in the y direction), leading to a localized rearrangement zone.

Spatial Periodicity of Rearrangement Zones. The periodicity of the rearrangement zones along the x direction can be explained in terms of the Read–Shockley model (27) of low-angle grain boundary. The tapered channel geometry caused a small misorientation (θ = 10°) between the crystal near the top wall and that near the bottom wall. A geometrically necessary dislocation array was required to accommodate this misorientation (Fig. 4; also see Movie S2). According to the Read–Shockley model, the misorientation here requires a dislocation array with Burgers vector $\boldsymbol{b} = (b_x = 0, b_y)$ and periodicity s, such that

$$\frac{b_y}{2s} = \tan\left(\frac{\theta}{2}\right). \quad [1]$$

A periodic dislocation array with the smallest possible Burgers vector that satisfies Eq. **1** is the configuration with the lowest energy for this grain boundary. The shortest Burgers vectors in the crystal did not point in the y direction, however. Instead, the two Burgers vectors in Fig. 3 *B* and *C* have $b_y = a\sqrt{3}/2$ and $b_x = \pm a/2$, where a is the lattice parameter which is approximately equal to one droplet diameter. As the dislocation shuttled between the walls, the x component of the Burgers vector alternated between $b_x = a/2$ and $b_x = -a/2$. The time-averaged Burgers vector was, therefore, purely in the y direction with $b_y = a\sqrt{3}/2$. The Read–Shockley model predicts the periodicity of the rearrangement zones along the channel to be $s = a\sqrt{3}/(4\tan(\theta/2)) = 4.9a$ (for $\theta/2 = 5°$), which is consistent with our observation: Fig. 5*A* shows that the average spacing s between the dislocations was 5.5 droplet diameters.

The Read–Shockley model also explains why the dislocation velocities in the x direction relative to the crystal nearly canceled the local extrusion velocity of the crystal. In this way, the x spacing between neighboring dislocations was maintained at the value s required by the model. Deviation from this ideal spacing would lead to a higher energy and a restoring force toward the ideal spacing. As our flow was in the quasi-static regime, the duration of a T1 was small compared with the rate at which the drops flowed (Fig. S1). The system was thus able to stay near the configuration with minimum energy. A careful look at the traces of x/a positions reveals that the x positions of the dislocations fluctuated (Fig. 5 *A* and *B*). Within one period of dislocation motion between the top and bottom walls, these traces were asymmetrical and the x position of the dislocations tended to drift to higher x values, or a more downstream location in the channel. This pattern indicates that the dislocations moved in a "stick–slip" fashion arising from the intrinsic lattice resistance (i.e., Peierls barrier) to dislocation motion (28).

Temporal Periodicity in T1 Motions. The most remarkable feature of the droplet dynamics here is the orderly, periodic motion of the dislocations in each rearrangement zone (Fig. 5*C*; also see Movie S2). Here, each time a dislocation reached the channel wall and escaped the crystal, a new dislocation was nucleated. The x component of the Burgers vector and the y component of the velocity vector of the new dislocation became opposite to the previous dislocation. This dislocation then traveled along the slip

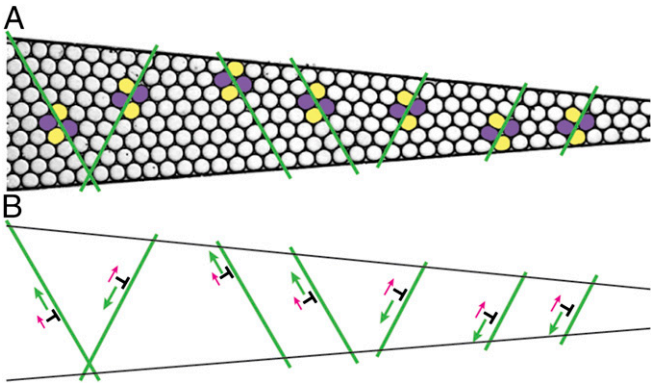

Fig. 4. Spatial distribution of dislocations and slip planes throughout the channel. (*A*) Microscope image of the emulsion. The slip planes are indicated by green lines. The dislocations and corresponding droplet pairs during the T1 are colored in yellow and purple. (*B*) The corresponding array of Burgers vectors (pink arrows) and their directions of motions (green arrows) along the respective slip planes (green lines).

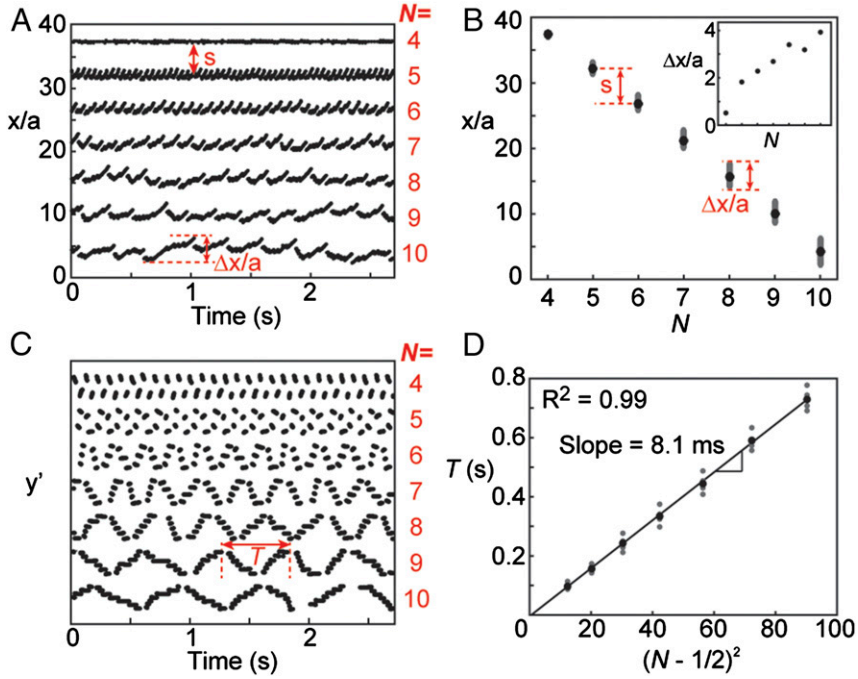

Fig. 5. Spatial and temporal periodicity of dislocation dynamics. (*A*) *x/a*-positions of T1s in the channel as a function of time. The mean spacing *s* between the seven sets of slips shown is $s = 5.5$. *a* is the diameter of one drop. Each trace represents the *x/a*-positions of the T1s within one rearrangement zone where *N* rows of drops reduces to $N - 1$. (*B*) Fluctuation in the *x/a*-position of the dislocations as a function of channel width expressed in terms of the number of rows of drops *N*. The height of the gray error bar represents the maximum fluctuation $\Delta x/a$ measured (also plotted in the inset). (*C*) Scaled *y* position (*y′*) of the dislocations as a function of time. For each trace, the maximum (or minimum) *y′* value represents the top (or bottom) of the channel wall within one rearrangement zone where *N* rows of drops reduces to $N - 1$. The *y* position of the T1s oscillates with period *T*. For both *A* and *C*, the positions plotted are that of the drops with five neighbors for convenience (*Materials and Methods*). (*D*) The period *T* scales linearly with $(N - 1/2)^2$. The periods are calculated using fast Fourier transform of the data in *C*; also see Fig. S2. Gray markers represent data from four independent experiments at the same flow conditions. Black markers represent the average from these experiments. The black line is the best linear fit to the averaged data.

plane with a constant velocity as required by the Read–Shockley model. The whole process subsequently repeated itself. Given that dislocation nucleation is usually a stochastic event, these predictable, repeatable nucleation events appeared surprising. The ordered nucleation events occurred here because of the confined geometry of our system: The tapered channel wall applied a compressive load on the extruding crystal, pressing the surface step which formed whenever the edge dislocation reached the channel wall and escaped the crystal. The surface step acted as a local stress concentrator and caused the next dislocation to nucleate at the same location spontaneously. This effect is analogous to heterogeneous dislocation nucleation at a surface step beneath an indenter (29). Because of this repeated heterogeneous nucleation of dislocations at transient surface steps, the sum of the number of dislocations and surface steps was conserved in the crystal being extruded through the channel. In the absence of dislocation interactions, the sequence of dislocation nucleation and glide events could repeat itself indefinitely with fixed temporal periodicities.

Based on the above discussions, the temporal period T of the dislocation motions can be predicted from the dislocation velocity. Consider a rearrangement zone where the number of rows of drops decreases from N to $N - 1$. At steady state, the volumetric flow rate Q is constant, and is related to the local flow velocity u_x by

$$Q = u_x\left(N - \frac{1}{2}\right)\frac{\sqrt{3}a}{2}H, \qquad [2]$$

where H is channel height. The x component of the dislocation velocity relative to the crystal is $v_x^{dis} = -u_x$. The y component of the dislocation velocity is $v_y^{dis} = \pm\sqrt{3}v_x^{dis}$. During period T, the dislocation travels a distance of $\left|v_y^{dis}\right| \cdot T = 2(N - (1/2))a\sqrt{3}/2$. Therefore, the period can be expressed as

$$T = \left(N - \frac{1}{2}\right)^2 \frac{\sqrt{3}a^2H}{2Q}. \qquad [3]$$

Indeed, Fig. 5*D* shows that T is proportional to $(N - (1/2))^2$. The slope of the linear fit is 8.1 ms, very close to the expected value of $(\sqrt{3}\,a^2H)/2Q = 7.6$ ms based on our experimental parameters.

Conditions for the Emergence of Order. An essential criterion of the ordered dynamics is that a new dislocation can only be nucleated at the transient surface step when an existing dislocation escapes the crystal. Once nucleated, the dislocation should move at an approximately constant speed so that the rearrangement zones stay well-separated. We now discuss briefly the conditions necessary for this order to emerge.

First, the Read–Shockley model requires the channel taper angle to be sufficiently small, so that the spacing s between neighboring rearrangement zones is sufficiently large to allow each dislocation to move relatively independent of each other. A large taper angle leads to a high density of dislocations, whose interaction could lead to various processes such as mutual blockage. In addition, if a dislocation's motion is blocked by another dislocation, new dislocations may be nucleated at locations other than the transient surface steps. All of these processes could lead to a chaotic behavior and disrupt the periodic dislocation dynamics (see Fig. S3 and Movie S4 for the case at a channel half-angle of 20°). Assuming a spacing of $s \sim 3$ droplet diameters is necessary to avoid dislocation interactions, the

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ordered dynamics should be observed almost indefinitely in tapers with a half-angle less than $\theta/2 \sim 8.2°$, as long as other criteria discussed below are satisfied. Indeed, at a half-angle of 8°, we started observing dislocation interactions (Fig. S4 and Movie S5). We note, however, that the disruptions caused by such interactions were transient. As the dot product of the Burgers vectors was positive in our system (Fig. 3), the dislocations repel each other rather than combine into junctions when they meet. The original order and periodicity were restored once the two interacting dislocations moved away from each other and exited the crystal, and new nucleations took place at the channel wall (Fig. S4). The ordered dynamical behavior here was thus surprisingly robust in this regard.

Second, the channel width needs to be sufficiently small. We observed that the fluctuation in the x position of the dislocation Δx increased with channel width (Fig. 5*B*). At a channel width of 10 droplet diameters, this fluctuation reached $\Delta x/a \sim 4$ at times. As the channel width increased further, $\Delta x/a$ could exceed the average spacing $s \sim 5$ between neighboring dislocations due to the stick–slip motion. It is then possible for dislocation trajectories to intersect and cause transient chaotic behavior. In our experiments, these intersections of trajectories started occurring at channel width exceeding ∼12 droplet diameters (Fig. S5). Such interactions are consistent with the lack of order in the dislocation dynamics in microscale crystals as the system size increases. We note, however, that in the absence of initial defects, we were able to observe order in channels with a width of up to 18 droplet diameters (Fig. S6 and Movie S6). The examination of an even larger number of drops is currently limited by the field of view and resolution of our imaging system.

Third, the flow velocity of the drops, or the extrusion speed of the crystal, needs to be sufficiently small. The dislocations have limited mobility due to the viscosity of the fluids. If the dislocations cannot move fast enough to accommodate the strain rate imposed by the extrusion speed, more dislocations will be nucleated without the assistance of surface steps. At high flow rates, the collection of drops behaves more like a fluid than a solid (30, 31). The frictional stress from the channel wall becomes large enough to activate slips on horizontal planes parallel to the x axis. Dislocation nucleation on horizontal slip planes was observed when the applied flow exceeded $Re \sim 10$ and $Ca \sim 10^{-2}$ at the constriction (Fig. S7). These slips can then interact with the ±60° slips and disrupt the periodic slip dynamics.

Fourth, the density of defects should be sufficiently low. Defects can arise from drops having sizes that deviate significantly from the others (corresponding to impurities in real crystals), or drops with the number of neighbors deviating from six (corresponding to interstitial or vacancies in real crystals) (Fig. S8 and Movie S7). These defects produce stress–strain fields that interact with dislocations, and can act both as dislocation obstacles and preferential sites for dislocation nucleation. Whereas atoms of the same species have identical size, the drops forming the soft crystal inevitably have different sizes due to the method with which they were generated and manipulated before their injection into the channel. In all our experiments, the drop size distribution was narrow (<3%) and the ordered dynamics was always observed as long as other conditions required for the order were satisfied. Nevertheless, we observed interstitial and vacancy defects in the crystal close to the entrance of the channel (Movie S8). Remarkably, these defects disappeared as the crystal flowed through the taper and reached a channel width of ∼12 droplet diameters. In other words, the extrusion of the crystal in the confined taper was able to anneal the crystal mechanically (32). Further research is needed, however, to elucidate the mechanism of annealing.

Conclusions

In summary, here we reported observation of ordered dislocation dynamics as a 2D soft crystal is extruded in a microfluidic channel. The spatial periodicity of the rearrangement zones is consistent with the Read–Shockley model of low-angle grain boundary. The temporary periodicity of dislocation dynamics is a result of repeated heterogeneous nucleation of dislocations at transient surface steps. The mechanism operates almost indefinitely as long as dislocations do not interact strongly. The system is also surprisingly robust; disruptions due to dislocation interactions are only transient and the original order is restored after the interactions end. This discovery is significant not only as a model for nanoscale extrusion of solids, but also for flow control in droplet microfluidic systems.

Materials and Methods

Droplet Generation, Emulsion Preparation, and Reinjection. We used methods in soft lithography to fabricate microchannels in poly(dimethylsiloxane) (33). The microchannels were rendered hydrophobic by treatment with Aquapel to avoid droplet wetting of the wall. We generated monodisperse 50-pL droplets using flow-focusing nozzles (34), where deionized water was the disperse phase, and a hydrofluoroether HFE-7500 (3M) was the continuous phase. The dimensions of the cross-section of the flow-focusing orifice were 40 μm × 30 μm (width × height), respectively. The dispersity of the drops was <3% by volume. The viscosity of HFE-7500 was ∼1.24 cP at 20 °C, the temperature at which all experiments were performed. The continuous phase contained an ammonium salt of Krytox (2% wt/wt) as surfactant to stabilize the drops against coalescence (35). We collected the drops generated from the flow-focusing nozzles in syringes (Normject 10 mL). As water has a lower density than HFE-7500 does (ρ = 1.63 g/mL), the drops creamed to the top of the syringes to form an emulsion with volume fraction $\varphi \sim 85\%$ after 12 h of storage at room temperature. The size of the drops remained unchanged after this storage time. Coarsening was insignificant in our system because: (*i*) coarsening due to the coalescence of droplets was eliminated as the drops were stabilized by surfactants against coalescence; (*ii*) coarsening due to Ostwald ripening was slow or insignificant. The monodisperse size distribution of the emulsion reduced the rate of ripening, a process driven by differences in the radius and the corresponding Laplace pressure of the drops.

Fig. 1*A* shows the geometry of a microchannel into which the concentrated emulsion was injected. The channel was tapered with a linearly decreasing width leading to a constriction downstream with a width of 30 μm. Unless otherwise stated, the half-angle of the taper was 5°. The channel had a height of 30 μm (in the out-of-plane direction). To drive the flow of the emulsion, we applied a constant pressure using a compressed air source, or a constant volumetric flow rate using a syringe pump. We found no difference in the flow profile of the emulsion using either method. We recorded movies of the drops flowing in the microchannel using an inverted optical microscope and a high-speed camera (Phantom v7.3). We used custom scripts written in MATLAB to track the position and the shape of all drops in each frame. More specifically, we converted each frame into a binary image, where the droplet border appeared as a black rim and the bulk of the drop appeared as a white region. We used built-in function *bwlabel* in MATLAB to identify and label all white regions which should correspond to all of the drops. Regions with areas smaller than 70% or larger than 130% of the mean droplet area were discarded. The *bwlabel* function outputs a label matrix having the same size as the input image, where each identified region was labeled with a unique integer. This information was then used as the input for the *regionprops* function, which outputs information including the centroid of each drop, and the position of the vertices.

Identification of T1 Rearrangement Events or Dislocations. We used the *voronoi* function in MATLAB to identify the number of neighbors for each drop in the emulsion. To facilitate the identification of T1 events or dislocations, we searched for drops that have five neighbors. We note that although this scheme does not identify T1 according to its formal definition given above, it was simple and correlated well with the occurrence of T1 events in our system as shown in Movie S2. The overall results as plotted in Fig. 5 did not change when the positions of drops with seven neighbors or the location between drops with five and seven neighbors were used.

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